2 – Evolutionary Robotics

Evolution, Genetics, Learning, GENITOR, Novelty Search

Evolutionary Robotics: Introduction

- Even simple biological systems are more autonomous than any artificial system
- The brain is structured at high level, but highly optimized and difficult to decode at network level
- The brain doesn’t use optimized learning rules like backpropagation, but things like Hebb’s rule

Evolutionary Robotics: Natural Evolution

- Most insects and other “simple” biological organisms hardly learn anything during their live

- The rely on **evolution** to adapt to environmental changes

- For more “complex” biological organisms that do learn (e.g. humans), the brain is structured for optimized learning (through evolution)

Evolutionary Robotics: Natural Evolution

- Darwin described 4 pillars of evolution:
  - **Population:** group of several individuals
  - **Diversity:** individuals have different characteristics
  - **Heredity:** characteristics are transferred over generations
  - **Selection:** individuals better at surviving will make more offspring

- **Genetics:** structure and operation of genes

- **Functional genomics:** role of genes in the organism

Evolutionary Robotics: Natural Evolution

- **Phenotype:** appearance, behaviour, etc. of the organism
  - Selection operates on the phenotype
  - The phenotype is affected by the environment, development, and learning

- **Genotype:** the genetic material of the organism
  - Selection does not directly apply on the genotype
  - The genotype is affected by mutations

Evolutionary Robotics: Natural Evolution

- An important thing to note is that evolution is only a concept
  - It does not think
  - It does not have a plan
  - It does not have any feelings
  - It is neither ‘bad’ or ‘good’

- But it is a very powerful concept!

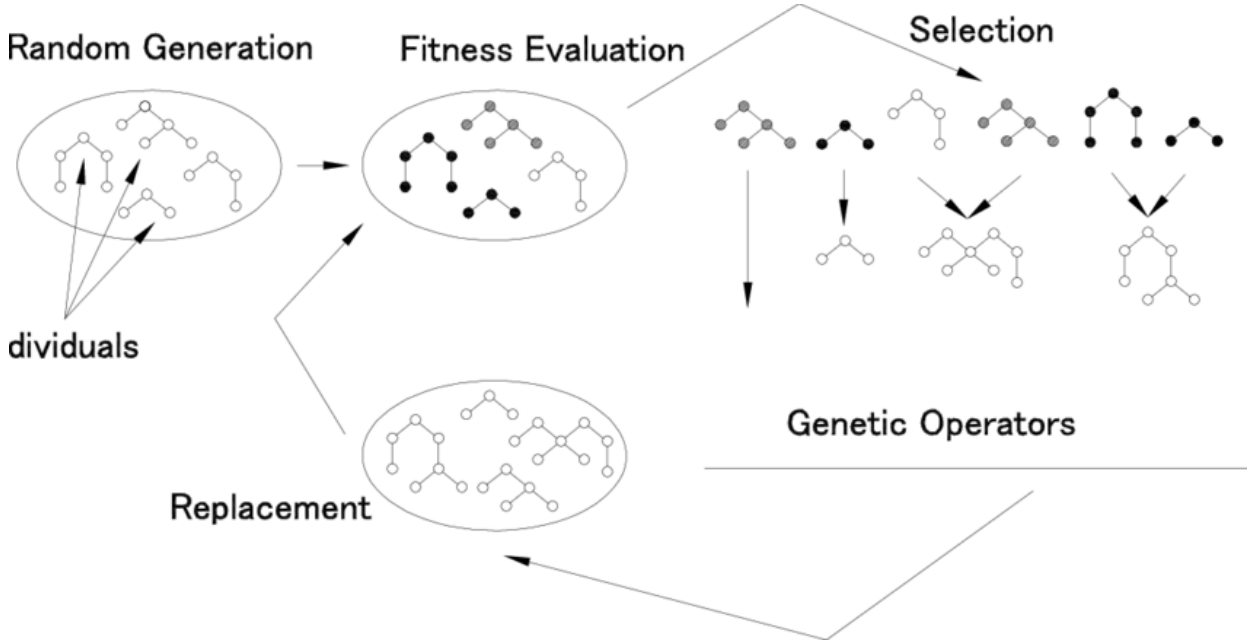
Evolutionary Robotics: Artificial Evolution

- **Artificial Evolution** is an automatic generation of solutions to hard problems

- Although similar in many ways to natural evolution; there are a few key differences:
  - **Fitness** is a performance measure for the individual solution
  - **Selection** is done according to a performance criterion (**fitness function**)
  - There is expected improvement between the initial and final solution

Evolutionary Robotics: Artificial Evolution

- **Evolutionary algorithms** are applicable to any problem domain if a suitable genetic representation, fitness, and genetic operators can be found



Evolutionary Robotics: Genetics

- The **Genetic Representation** describes the elements of the genotype and the mapping to the corresponding phenotype
- Must allow for genetic operators
- Set of possible genotypes should contain the optimal solution
- The choice of representation is helped by **domain knowledge**

Evolutionary Robotics: Genetics

- **Discrete Representation:** a sequence of (independent) discrete values (e.g. bitstring)

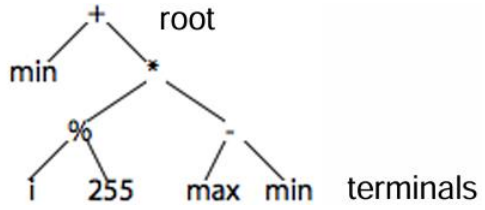
- **Sequence Representation:** mostly used for TSP-like problems

- **Real-valued Representation:** sequence of numeric parameters

- **Tree-based Representation:** used to encode computer functions

Expression r=min+(i/255)(max-min)

Nested list (+, min, (*, (/, i, 255), (-, max, min)))



Evolutionary Robotics: Initial Population

- The initial population should be sufficiently large and diverse to cover the entire problem space (usually 10-1000 individuals)
- Usually initialized as uniform samples over the solution space
- Be careful with bootstrapping! Previously evolved or handcrafted genotypes may be biased to the point of no return

Evolutionary Robotics: Fitness Function

- Evaluate the performance of the phenotype with a numeric score
- A fitness function is an example of **objective fitness**; you get what you evaluate
- **Subjective fitness:** hand-pick individuals if fitness cannot easily be quantified objectively

Evolutionary Robotics: Fitness Function

- The choice of fitness has a huge impact on the success of an EA
  - Normalize numbers correctly
  - Avoid discontinuous functions
  - Reduce the number of local optima

- **Multi-Objective Optimization:** two or more conflicting fitness functions
  - No single optimal solution, but a set of **Pareto-optimal solutions**
  - **Pareto Front:** set of solutions that do not dominate each other, but do dominate any solution not in the Pareto front

Evolutionary Robotics: Selection

- Make sure that the better individuals produce more offspring

- **Selection pressure** is inversely proportional to the number of selected individuals
  - A high selection pressure leads to a loss of diversity and premature convergence
  - A low selection pressure leads to slow convergence

Evolutionary Robotics: Selection

- **Proportionate Selection:** the probability that an individual produces offspring is proportionate to its fitness (compared to the rest of the population)
  - Often known as **roulette wheel selection**
  - Problems: difficult to control selection pressure
- **Rank-based Selection:** assign individuals a rank based on their fitness, use this to select individuals (through e.g. roulette wheel)
- **Truncated Rank-based Selection:** only the top *X* individuals are allowed to make offspring (and each one makes the same amount, e.g. out of 100 individuals make 5 copies of the top 20)

Evolutionary Robotics: Selection

- **Tournament Selection:** for every offspring to be selected:
  1. Pick *k* individuals from the population
  2. Choose the individual with the highest fitness
  3. Put all individuals back in the population

- Larger *k* = Larger selection pressure

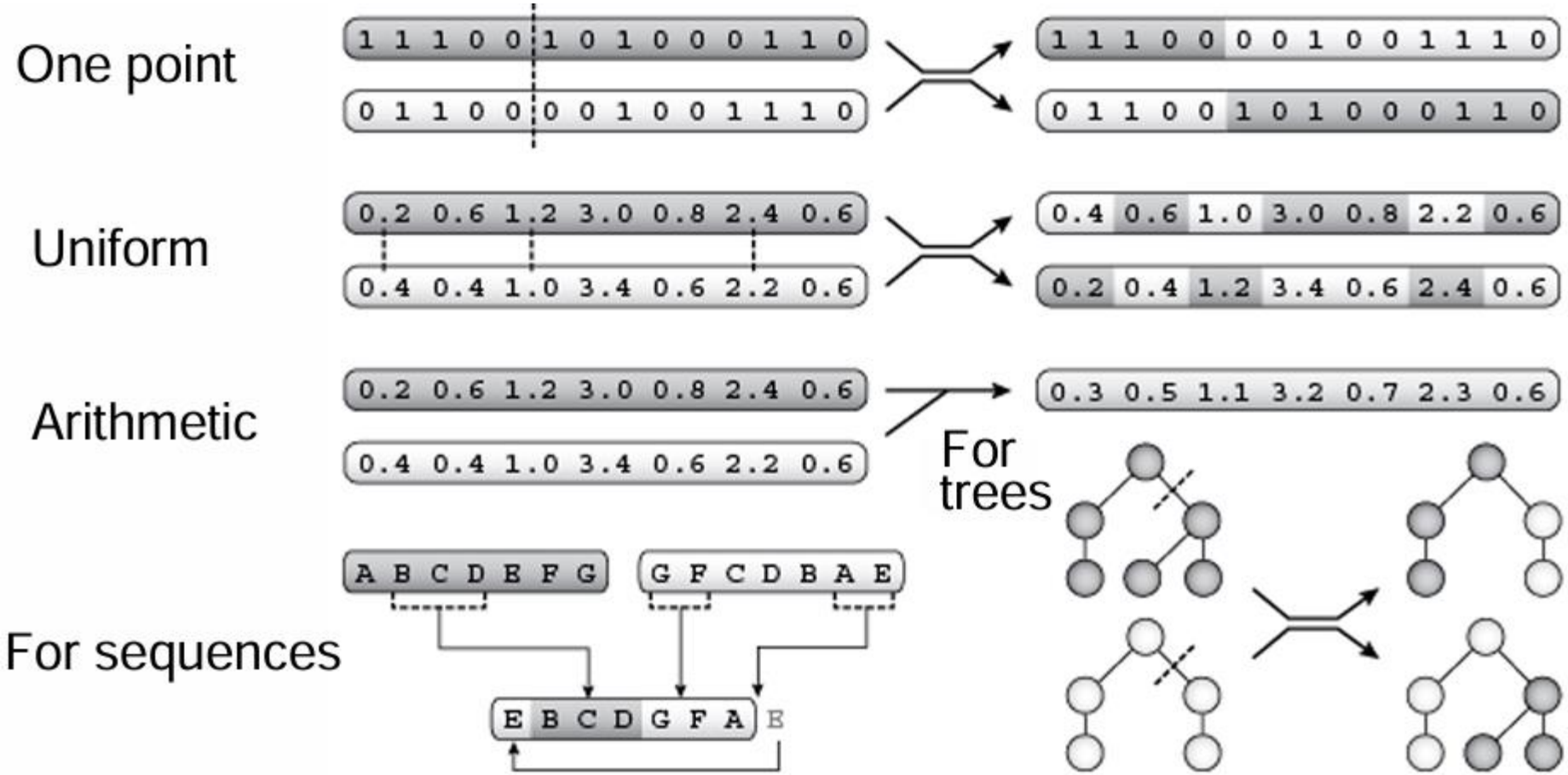
Evolutionary Robotics: Reproduction

- **Generational Replacement:** old population is entirely replaced by offspring (most frequent in practise)

- **Elitism:** maintain the *n* best individuals from the previous generation to prevent loss due to mutations

- **Generational Rollover:** place offspring at the positions of the worst performing individuals

Evolutionary Robotics: Crossover



Evolutionary Robotics: Mutation

Binary genotypes

1 1 1 0 0 1 0 1 0 0 0 1 1 0

1 1 0 0 0 1 0 1 1 0 0 1 1 0

Real-valued genotypes

0.2 0.6 1.2 3.0 0.8 2.4 0.6

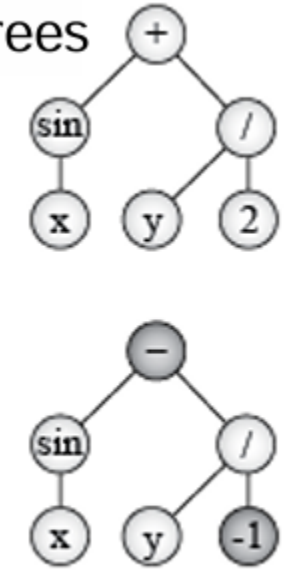
0.2 0.7 1.2 3.0 0.8 2.2 0.6

Sequence genotypes

G F C D B A E

G F C E B A D

For trees



Evolutionary Robotics: Artificial Evolution

- How to estimate the **fitness landscape**:
  1. Sample random genotypes (flatter landscape = more samples)
  2. Apply genetic operators for a fixed number of timesteps
  3. The larger the fitness improvement, the easier it is to evolve

- Fitness graphs are only useful if the problem is stationary!
- Stagnation may mean optimality or premature convergence

- **Diversity** of population shows if there is room for further evolution

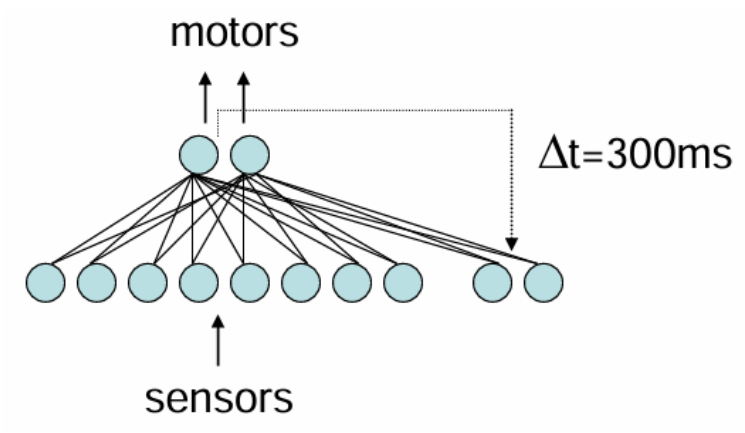
Evolutionary Robotics: Evolutionary Robotics

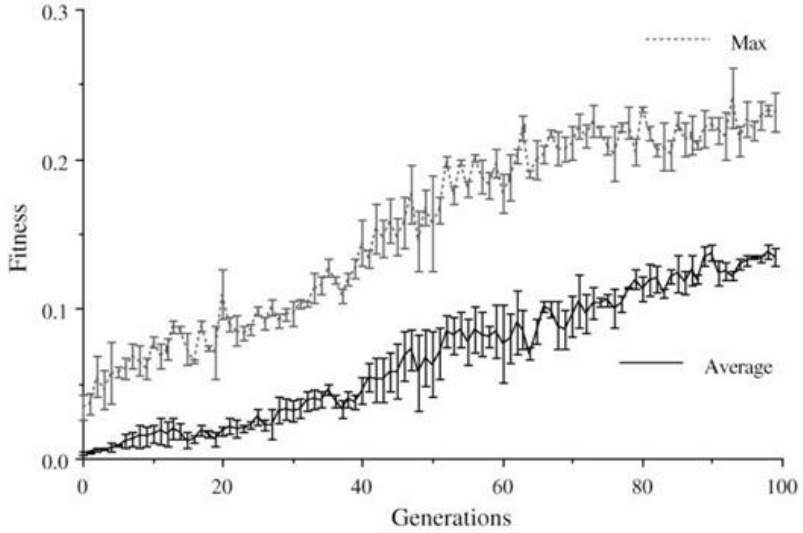
- **Evolutionary Robotics:** the automatic generation of morphology and control of autonomous robots

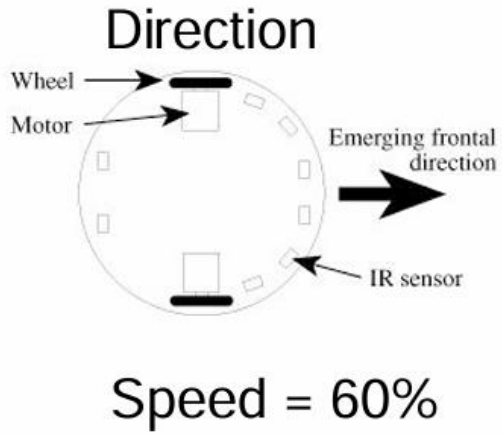
- Motivation:
  - Interaction between robot and environment is very complex and hard to predict; difficult to design from a purely top-down perspective
  - It is a synthetic, rather than analytic, approach to the study of adaptive behaviour; evolved robots will sometimes develop behaviours that humans never thought of

Evolutionary Robotics: Example

- Autonomous, collision free navigation of a simple differential drive mobile robot:



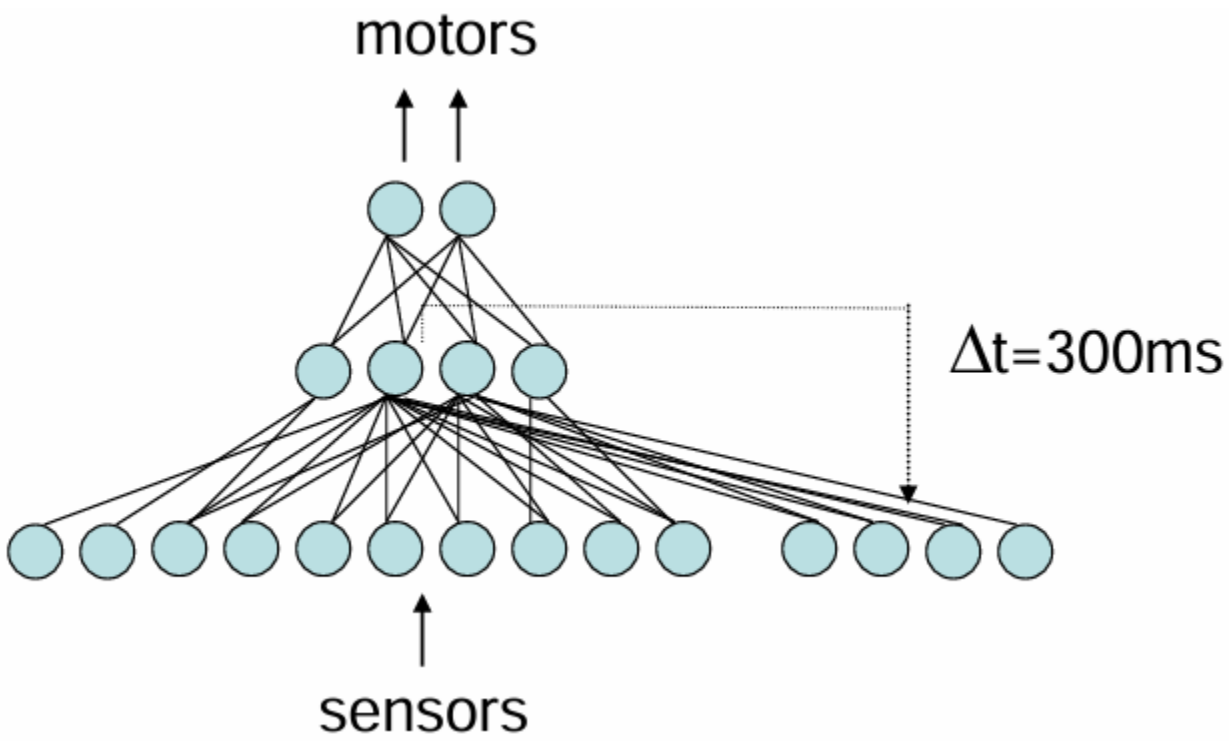




- Fitness function: $V \cdot \left(1 - \sqrt{\Delta v}\right) \cdot (1 - i)$

Evolutionary Robotics: Example

- For larger networks, use the hidden layer as feedback



Evolutionary Robotics: Evolution & Learning

- Evolution leads to very slow adaptions (requires at least one generation), might be too slow for quick changes
- Instead of evolving the weights of the neural network, evolve the rules and parameters of the training / learning of behaviour
- This results in slow adaption through evolution, but fast adaptation through continuous learning

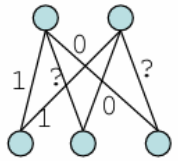
Baldwin effect: Learned behaviour has an influence on the chance to reproduce

Mechanism	Time scale	Affects
Evolution	Slow, adaptation to changes over several generations	Genome
Learning	Medium, adaptation over life of single individual	Phenotype, weights of neural network
Short-term memory	Very fast, can be immediate response to encountered situation	Phenotype, state of (some) neurons in neural network

The control loop in the previous examples is an example of short-term memory

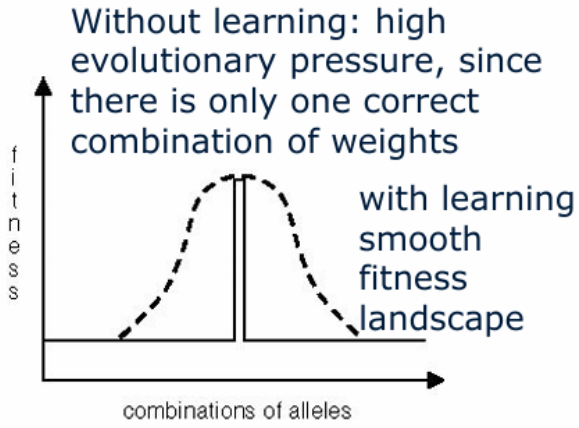
Evolutionary Robotics: Evolution & Learning

- **Hinton & Nowlan Model:** leave some genes to be learned

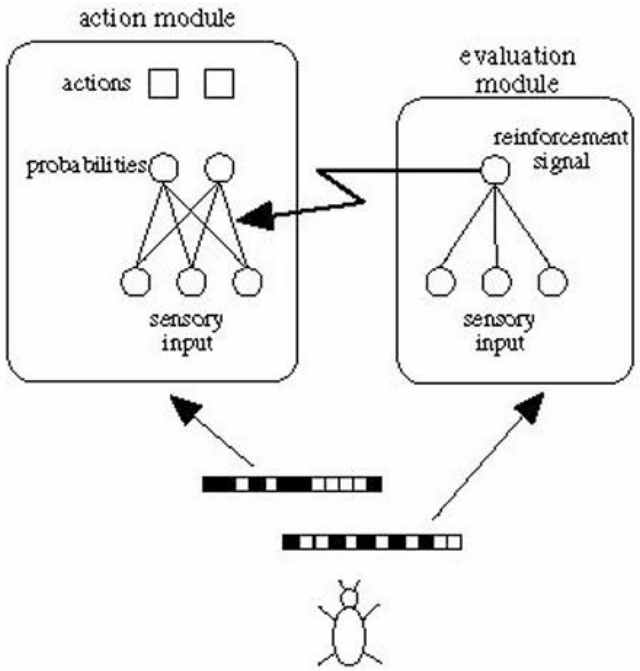


00?11???0111?0?1?0?1

Genes = weights are determined by evolution (0 or 1) or selected by evolution to be adjustable through learning (?)
Fitness=correct combination of weights

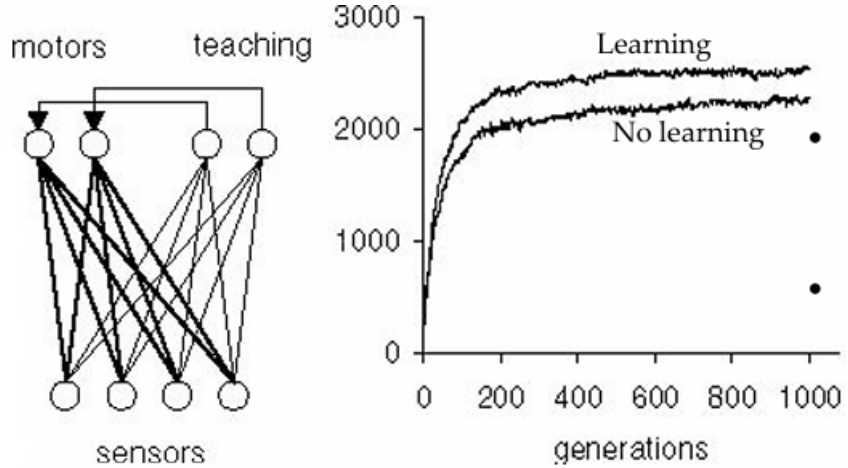


- **Evolutionary Reinforcement Learning:**
  - Leads to better performance than just evolution or reinforcement learning alone!



Evolutionary Robotics: Evolution & Learning

- **Evolutionary Auto-Teaching:** one half teaches the other one at runtime using the Delta rule (allows for dynamic environments)



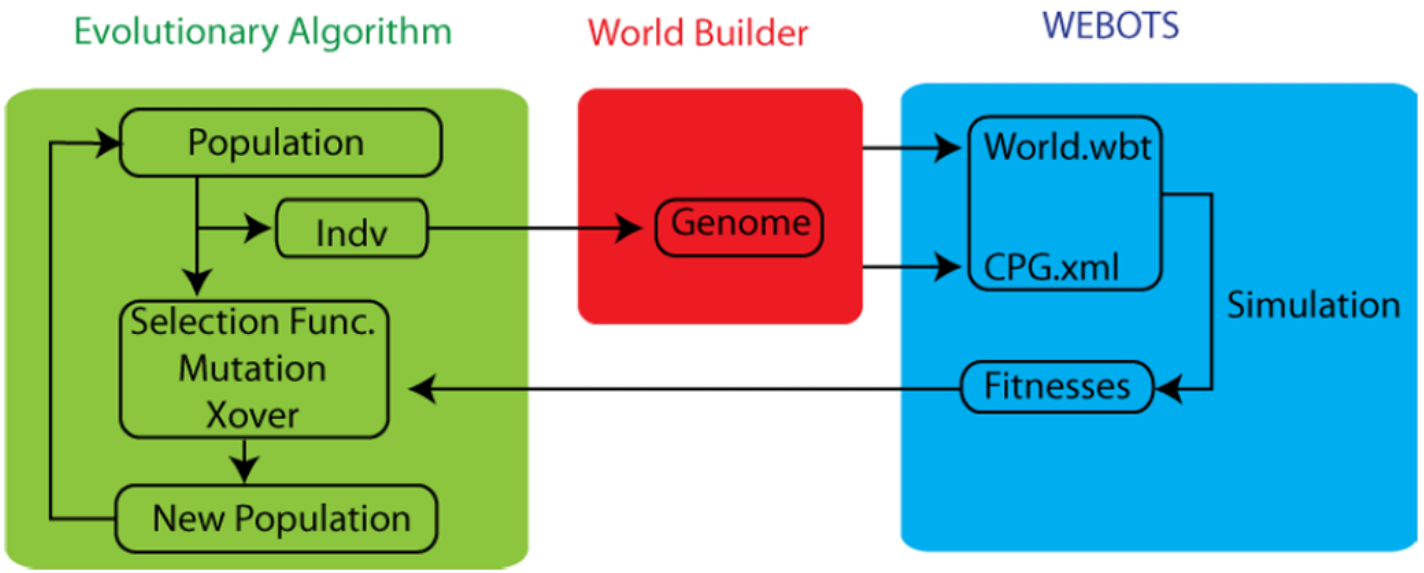
- Evolved adaptive individuals can cope with changes in the environment, and tend to generalize better to the real world

Evolutionary Robotics: Evolution & Learning

Evolution	Learning rules
Both: mechanisms for adaptation Both: adaption of control and behavior	
Slow adaptation	(Relatively) fast adaptation
Adaptation of morphological parameter	Typically only adaptation of control
Guided by environment	Often guided by single subject
Supports adaptation on population level, helps a species to survived	Adaptation of individual subject, helps subject to survive

Evolutionary Robotics: Co-Evolution

- Both morphology and control have to suit the same task
  - **Morphology** can be represented as a directed graph
  - **Control** can be represented as a neural network



Evolutionary Robotics: GENITOR

- **GEN**etic **I**mplemen**TOR:**

1. Create initial population
2. Evaluate fitness and rank of each solution
3. Repeat:
    1. Select two parents
    2. Create new solution from parents
    3. Evaluate fitness of child
    4. If fitness > minimum fitness in the population, insert it at the appropriate rank and remove the worst solution

Evolutionary Robotics: GENITOR

- Has a fast reaction time, since it does not need to wait for generational replacement
- Built-in elitism without needing to decide the parameter
- Maintains diversity
- Mainly a good foundation for other algorithms

Evolutionary Robotics: Coevolutionary Comp.

- **Coevolutionary Computation:** evolve the problem and solution together as 2 separate populations (usually using GENITOR)

- Each population has its own fitness, the 2 populations constantly interact and coevolve

- Fitness of solution depends on the last *x* problems it encountered
- Fitness of problem depends on the last *x* solutions it encountered
  - Predator-Prey relationship

Evolutionary Robotics: Novelty Search

- Objective functions may unintentionally misdirect to dead-ends
  - E.g. a maze where fitness = distance to exit

- Instead of seeking an objective, seek **behavioural novelty**

- For example, a biped robot learning to walk
  - First robot will fall (remember this behaviour)
  - To obtain additional rewards, robot must learn how to not fall (i.e. walk)
  - To obtain additional rewards, robot must learn how to walk further, etc.

Evolutionary Robotics: Novelty Search

- Novelty search only works if the measure of novelty is appropriately defined

- **Fitness-Diversity Driven Coevolution (FDDC):** use diversity in fitness to define novelty (skips having to keep track of the history of behaviours)
- Another approach is to use the sparseness of the solutions in the solution space (denser solutions = less novelty)